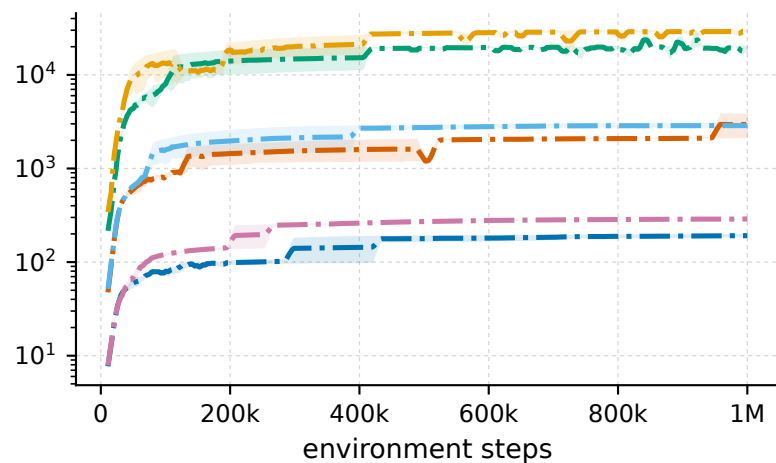
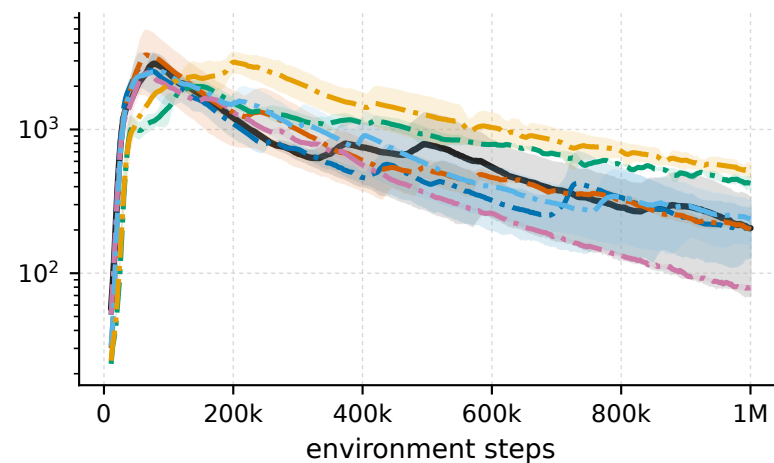


HumanoidStandup-v5 - FHR + SAC internals (mean $\pm$ 1 s.e.m., 600-point binning)

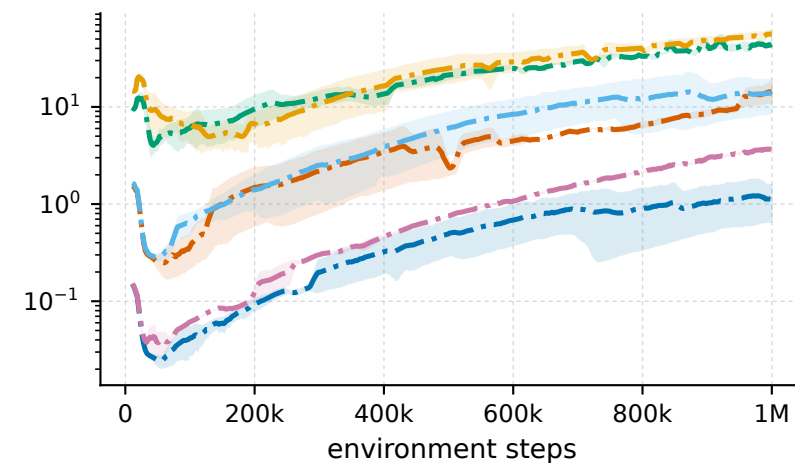
$\lambda \cdot \text{penalty}$ (weighted)



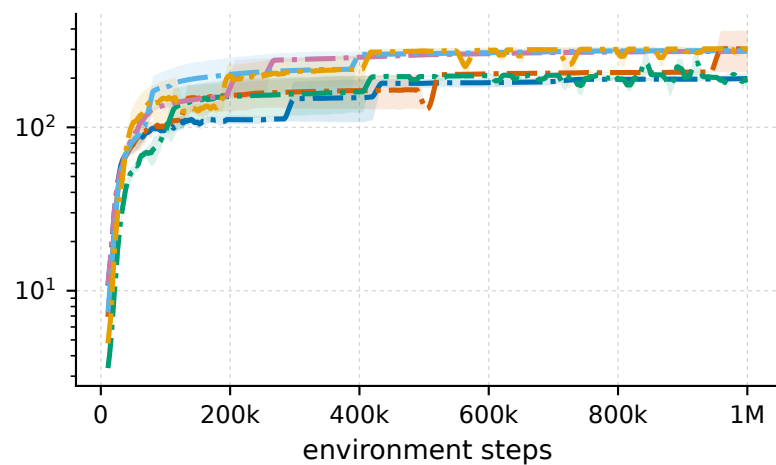
critic TD loss



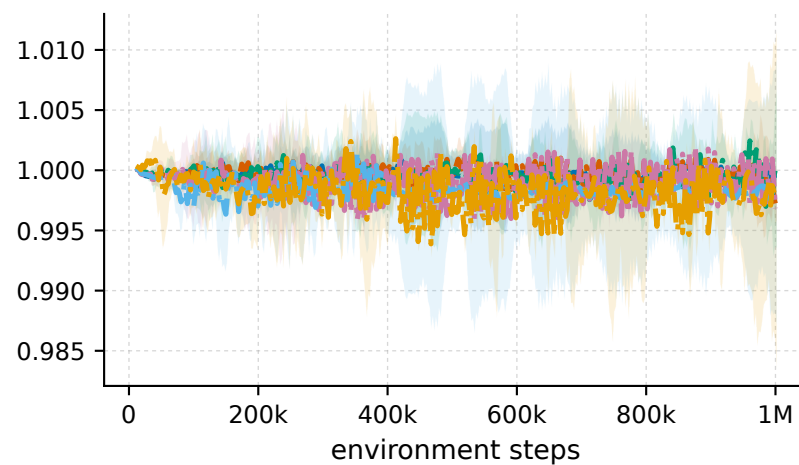
$\rho = \lambda \cdot \text{penalty} / \text{TD}$



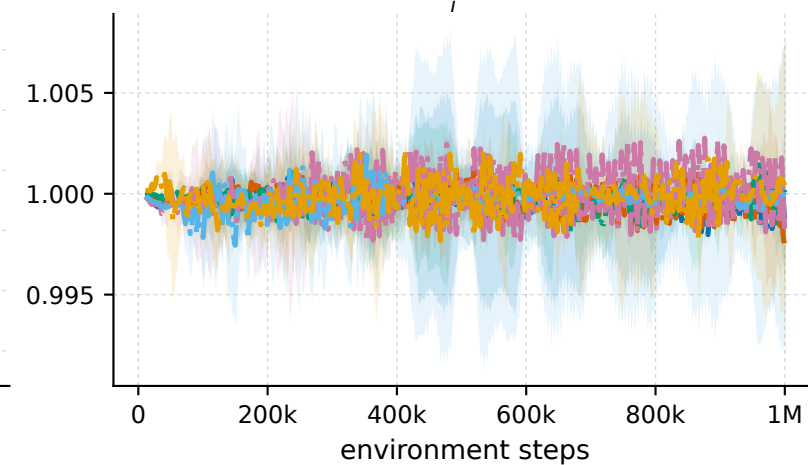
recurrence residual RMS



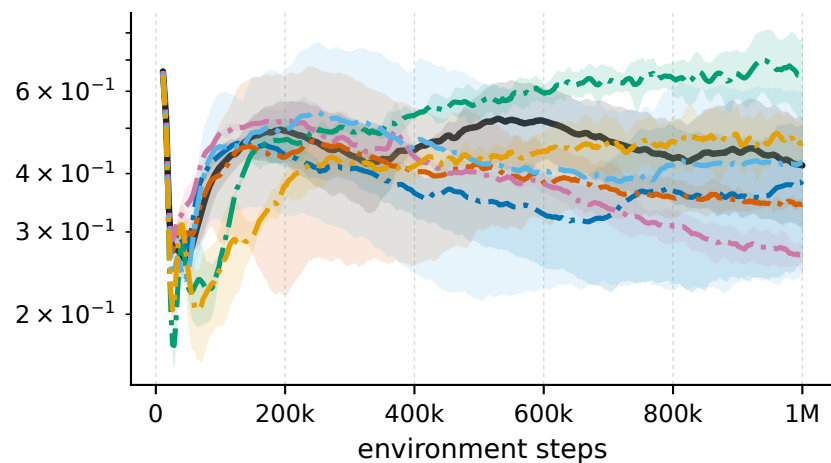
companion spectral radius



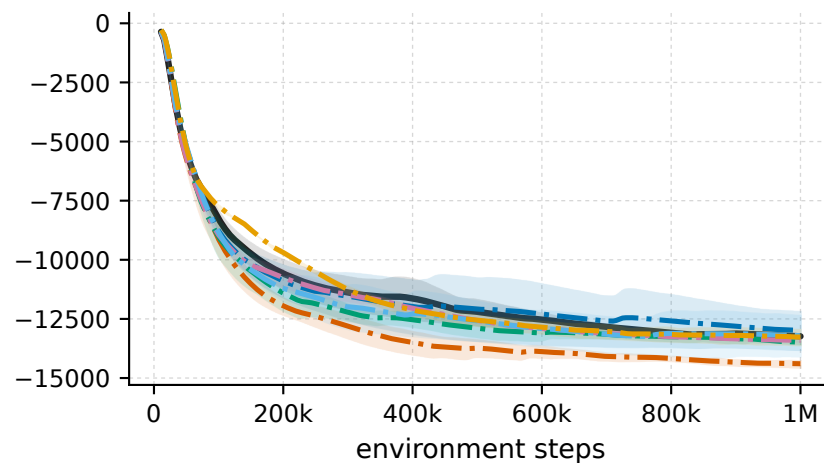
$\sum_i c_i$



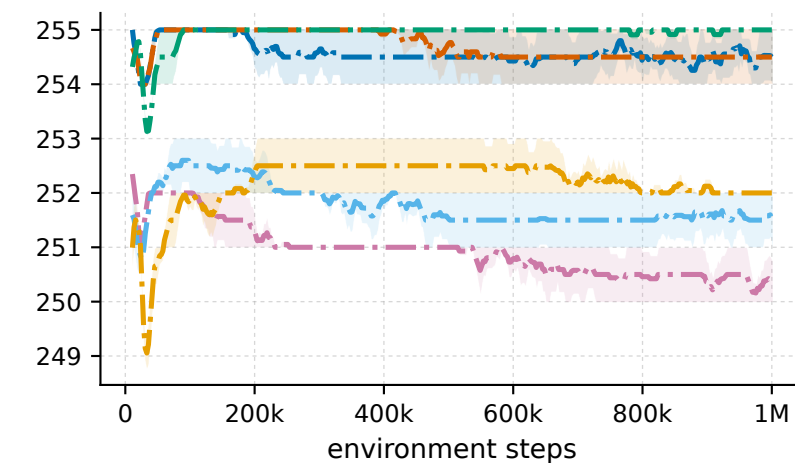
temperature α



actor loss



penalty batch size



— baseline (stock SB3 SAC)
— FHR $\lambda=1$, $r=4$ (learned c , + PER)
— FHR $\lambda=10$, $r=4$ (learned c , + PER)

— FHR $\lambda=100$, $r=4$ (learned c , + PER)
— FHR $\lambda=1$, $r=12$ (learned c , + PER)

— FHR $\lambda=10$, $r=12$ (learned c , + PER)
— FHR $\lambda=100$, $r=12$ (learned c , + PER)